

# Refrigerator Ingredient Recognition

Vaishnavi Borse  
7075846

Kaixin Zhang  
7076400

## 1. Task and Motivation

### Task Description

We aim to investigate and compare three different approaches for object recognition in refrigerator scenes: The goal is to identify and label food items inside refrigerator environments using computer vision techniques.

- **YOLO (You Only Look Once):** Firstly, YOLO alone then YOLO followed by SAM for mask refinement. A real-time object detection algorithm that identifies and classifies visible ingredients inside a refrigerator.
- **SAM (Segment Anything Model):** SAM followed by CLIP for zero-shot classification. A segmentation model that extracts fine-grained object masks to separate overlapping or partially visible items.

The workflow begins with capturing images of refrigerator contents using a smartphone. YOLO is applied to detect known ingredient classes, while SAM refines the segmentation of individual items, improving detection accuracy in cluttered or occluded environments. The recognized ingredients.

### Motivation

Object recognition in structured, cluttered, and constrained environments such as the interior of a refrigerator is a key enabler for smart kitchen systems, dietary monitoring, and personalized food recommendation engines. Accurate identification of stored food items supports automation, reduces food waste, improves nutritional awareness, and enables intelligent culinary assistance.

However, typical object detection models trained on general-purpose datasets like COCO or PASCAL VOC struggle in such specialized settings. Refrigerator environments introduce several unique challenges:

- **Cluttered Layouts:** Items are densely packed, often overlapping or stacked, leading to frequent occlusions and partial visibility.
- **Constrained Space:** Tight, non-standardized interiors restrict camera placement and viewing angles.

- **Lighting Variability:** Dim or uneven lighting and reflective surfaces (e.g., plastic or glass containers) reduce image clarity and consistency.
- **Visual Similarity:** Many food items (e.g., dairy products, beverages) have similar packaging, complicating fine grained classification.

These conditions significantly degrade the accuracy and reliability of conventional object detection pipelines. Therefore, there is a strong need for tailored computer vision solutions that can effectively handle dense layouts, occlusions, and challenging visual conditions. Addressing these issues is not only of academic interest but also vital for developing robust, real world intelligent food management systems that support health aware and sustainable lifestyles.

## 2. Goals

- **Mid-term Milestones:** The mid-term goal is to establish a working prototype of the system by achieving the following:

Successfully train a YOLOv8 object detection model on a labeled refrigerator dataset, ensuring it can identify key food items despite clutter and occlusion.

Integrate the Segment Anything Model (SAM) to refine object boundaries using YOLO predicted bounding boxes as priors. This hybrid approach is expected to improve segmentation precision in tightly packed scenes.

Generate and analyze preliminary evaluation metrics including mean Average Precision (mAP) for detection, and Average Recall at k (AR@k) for segmentation performance. These metrics will be reported for all three stages: YOLO-only, YOLO+SAM to compare their individual and combined effectiveness.

- **To Address the Challenges:** The project aims to tackle core difficulties inherent to fridge environments, including high object density, poor lighting, and frequent occlusions. These factors often result in lower detection and segmentation accuracy using standard

models. The focus will be on evaluating how the combination of detection (YOLO), segmentation (SAM), and semantic classification (CLIP) can improve robustness in such challenging scenarios. Additionally, trade-offs will be assessed in terms of model speed (real-time capability), segmentation precision (mask quality and IoU), and classification robustness (especially under ambiguous or unseen objects). This comparative study will help in choosing the most balanced approach for deployment in smart kitchen or dietary monitoring systems.

### 3. Models and Methodology

#### Frameworks Used

- **YOLOv8 (Ultralytics):** Employed for object detection, YOLOv8 provides a fast and robust baseline for identifying food items inside refrigerator environments.
- **Segment Anything Model (SAM):** Used for instance segmentation, SAM produces high-quality object masks that refine YOLO's coarse bounding boxes or act as independent region proposals.

#### Method Justification

- **YOLOv8** is selected for its speed and accuracy trade-off, especially in real-time scenarios. It serves as a solid backbone for detecting food items in dense and cluttered fridge layouts.
- **SAM** is incorporated to generate fine-grained instance masks, improving spatial understanding and enabling better isolation of overlapping objects.

#### Comparison with Existing Approaches

- Unlike traditional end-to-end object detection pipelines, the proposed approach decouples the detection, segmentation, and classification stages. This modularity enables the exploration of hybrid strategies and domain-specific optimization.
- Most existing methods are benchmarked on general-purpose datasets (e.g., COCO) and do not address the constraints posed by refrigerator environments, such as occlusion, lighting variance, and object similarity. Our approach specifically targets these real-world challenges.

#### Computational Budget

- All models rely on publicly available pre-trained backbones to reduce computational cost and training overhead. Specifically, we use:

- YOLOv8s or YOLOv12s for lightweight and efficient detection,
- SAM with ViT-B backbone for instance segmentation,
- Inference and experimentation are performed using a single NVIDIA A100 GPU ( on Google Colab ). No full end-to-end training is performed for SAM or CLIP, as their zero-shot or frozen capabilities are sufficient for our pipeline.

### 4. Datasets

Public fridge object detection dataset (3000 labeled images with 30 object classes). Additional 50-100 custom images captured under realistic conditions (occlusion, shadow, overlapping items) for generalization testing. The dataset is chosen for its relevance to structured, cluttered scenes, and real-world kitchen applications.

### 5. Evaluation

Evaluation Metrics: - Mean Average Precision (mAP@0.5), Average Recall at k (AR@1, AR@10, AR@100), Top-1 classification accuracy. Inference time per image (for runtime analysis) We will not define custom metrics but will conduct qualitative visual comparisons for segmentation precision and classification errors.

### References

Redmon et al., 2016. YOLOv1: You Only Look Once. Jocher et al., 2023. YOLOv8 by Ultralytics. <https://github.com/ultralytics/ultralytics> Kirillov et al., 2023. Segment Anything. Meta AI. Radford et al., 2021. CLIP: Connecting Text and Images. OpenAI. Li et al., 2022. BLIP: Bootstrapped Language-Image Pretraining. Salesforce Research. Chen et al., 2022. Hybrid object detection pipelines. CVPR. Liu et al., 2023. Dense object layout detection. ECCV.

Related Work: YOLO-based models have been widely adopted for real-time object detection due to their speed and accuracy (Redmon et al., 2016; Jocher et al., 2023). Segment Anything Model (SAM) by Kirillov et al. (2023) enables high-quality zero-shot segmentation using flexible prompts. OpenAI's CLIP (Radford et al., 2021) and Salesforce's BLIP (Li et al., 2022) allow zero-shot image-to-text matching and classification. Some recent studies (e.g., Chen et al., 2022; Liu et al., 2023) have explored multi-stage pipelines combining detection and segmentation, though few focus specifically on structured kitchen environments.